FeelMySparks 748i

Study-level development roadmap · FBW and aircraft systems

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Ambition: Build a demanding, coherent, and well-documented study-level simulation of a 747-8I—not merely a responsive control feel. The aircraft must connect its flight controls, hydraulics, autoflight, warnings, failures, procedures, and performance in a testable way. “Study-level” is a project target, not a claim of certification or perfect Boeing fidelity.

Where the project is now

The aircraft is based on the Sparky744 747-400 project. Existing hydraulic, autopilot, and other inherited systems remain the baseline. The FMC display has custom alert colors and transparent backing. The standalone custom Weight & Balance plugin was removed.

The new FBW work is a **partial roll-focused augmentation** integrated as an isolated Lua module in the existing XTLua hydraulic mixer. Law state is sampled at 20 Hz; the hydraulic model applies the result in its normal update. Current features are speed-scheduled outer-aileron roll response, flap-linked outboard-aileron droop, prototype load-factor MLA through outboard aileron/selected spoilers, capped manual-flight roll-rate damping, AP-servo bypass for that damping, and a native force-feedback stick-shaker warning cue where supported.

The user reports that the current handling is working well but feels slow; the 20 Hz increase is the latest responsiveness iteration and should be assessed separately in the next flight tests.

Critical distinction: protection versus warning

The stick shaker is a **warning cue**. It is not airframe buffet and it is not the protection law. A protection law would have to act earlier, based on valid and sufficiently independent measurements, and alter the appropriate control commands to keep the simulated aircraft within its specified envelope. Our current module does not command elevator or rudder and does not prevent an angle-of-attack/stall excursion. It therefore does not currently provide stall protection.

The list of missing features below describes gaps in **our module and project plan**; it is not a claim that a real 747-8 lacks or includes those systems. Before calling any law “747-8 accurate,” verify the actual aircraft behavior and architecture using authorized, credible technical references. Where exact data is unavailable, label the implementation as a simulator approximation.

Definition of “full study-level” for this project

- Every modeled function has a written requirement, a known owner, inputs/outputs, modes, limits, failure behavior, and cockpit indication.
- Normal, abnormal, and failure states are coherent across flight controls, hydraulics, electrical power, sensors, autoflight, warnings, and procedures.
- Actions have observable consequences; annunciations and checklists agree with the simulated state.
- Critical behavior has reproducible tests across weight, center of gravity, speed, altitude, configuration, and failures.
- Inherited systems are documented as inherited, modified, replaced, or not modeled; no behavior is silently double-applied.
- Verified source-based behavior is distinguished from educated approximation and tuning preference.

FBW target architecture

1. **Evidence and requirements:** Build a source-linked behavior matrix. For each proposed law, capture the source, aircraft/variant applicability, operating modes, control effect, protections, cockpit cue, failure response, and confidence. Do not treat forum summaries as design authority.
2. **Signal and actuator model:** Inventory X-Plane inputs, units, update rates, valid ranges, aircraft-specific DataRefs, hydraulic actuators, control-surface ownership, and AP inputs. Add explicit validity, filtering, rate limits, saturation, and deterministic reset behavior.
3. **Mode and channel management:** Define normal and reversionary modes, channel health, monitoring, disagreement detection, transfer/engagement logic, and unambiguous cockpit indications. Do not invent real 747 redundancy details without evidence.
4. **Control laws:** Design and test pitch, roll, and yaw command paths only after the aircraft-specific laws and sign conventions are understood. Include trim coordination and control feel consistently with the intended hybrid architecture.
5. **Envelope functions:** Specify speed/angle-of-attack/load/attitude/configuration boundaries and control priorities from evidence. Candidate topics to investigate include low-speed/stall margin, overspeed, load-factor, bank/attitude, configuration, and tail-strike protection. Implement only confirmed behavior or explicitly labeled simulation approximations.
6. **Load and disturbance functions:** Treat MLA, spoiler/aileron coordination, gust-load alleviation, and any OAMS-like function as separate, measurable laws. OAMS should not be represented by generic roll damping; model it only if reliable 747-8-specific evidence supports it.
7. **Autoflight integration:** Test AP/FD commands through the same control authority, respecting AP modes, servo engagement, trim, envelope limits,

handoff, disconnect, and failure cues. Prevent FBW damping or limiters from quietly fighting AP commands.

8 **Warnings and diagnostics:** Keep shaker, aural/visual warnings, and protection interventions distinct. Display the active mode/protection and log sensor, command, actuator, and limit states for debugging.

FBW implementation roadmap

Phase 0 · Baseline. Freeze a known-good aircraft copy; log control ratios, surfaces, AP state, speeds, load, configuration, and hydraulic pressure. Verify master-off equivalence. **Exit:** repeatable baseline and rollback path; no unexplained control ownership.

Phase 1 · Current roll laws. Evaluate 20 Hz scheduling, manual roll, AP turns, droop, MLA, spoiler coordination, high-speed lockout, and turbulence. Tune one factor at a time. **Exit:** no oscillation, sign reversal, AP conflict, or configuration-specific surprise in the test matrix.

Phase 2 · Requirements and modes. Document supported sources and mode transitions; define data validity, failure annunciations, reset/reversion behavior, and control-authority priorities. **Exit:** every requirement traces to a source or is clearly tagged as an approximation.

Phase 3 · Pitch/trim path. Map inherited elevator, stabilizer trim, AP pitch command, and hydraulic actuators. Prototype pitch augmentation in a separate opt-in branch, starting with trim/command coordination rather than a hidden hard guard. **Exit:** stable manual/AP behavior and safe disable/reversion demonstrated over the test matrix.

Phase 4 · Envelope protection. Implement only source-supported or clearly labeled protections; separately test normal law, limit engagement/release, sensor failure, and pilot override/reversion. **Exit:** protection acts before the defined boundary, releases correctly, and has clear cockpit state; no false safety claims.

Phase 5 · Yaw and disturbances. Audit rudder, yaw damper, engine-out behavior, spoilers, gust sensing, and any OAMS proposal; avoid duplicating existing X-Plane/aircraft functions. **Exit:** correct direction, bounded authority, AP compatibility, and graceful behavior through failures.

Phase 6 · Failures and validation. Inject sensor, actuator, hydraulic, electrical, and channel failures. Add scripted math tests, simulator scenarios, telemetry review, and regression checks. **Exit:** documented degraded modes; all critical tests repeatable and pass after changes.

Immediate next work

1 Test the new 20 Hz cadence against the user-tested prior feel. Record IAS, weight/CG, flap setting, bank target, AP state, turbulence, roll rate, and surface commands.

2 Verify roll-response curve, differential signs, AP damping bypass, MLA onset/release, droop, high-speed lockout, and FBW-master-off behavior at

safe altitude.

3 Build a feature/source matrix for actual 747-8I flight-control behavior; mark each item “verified,” “unverified,” or “project approximation.”

4 Map inherited pitch/trim/AP/hydraulic ownership before writing any elevator law. Do not add a stall guard based only on shaker behavior or intuition.

5 Choose the first pitch-law prototype from the evidence matrix; add telemetry and tests before making it a default.

Study-level aircraft beyond FBW

Full study-level quality is an aircraft-wide systems goal. These are workstreams to audit against the actual package; they are not assertions that every system is absent.

- **Flight controls / hydraulics:** Surface authority, trim, control feel, hydraulic sources, pressure/quantity/leak failures, redundancy, reversion, indications, and AP/FBW interactions.
- **AFDS / FMS / navigation:** Mode arming/capture/transition, VNAV/LNAV and approach behavior, tuning, constraints, route discontinuities, mode annunciation, disconnects, and edge cases. Preserve the working FMC while testing.
- **Engines / fuel:** Variant-appropriate engine model and limits, start/shutdown, thrust response, fuel feed/transfer, balancing, low fuel, pump/valve failures, and engine-out effects.
- **Electrical / hydraulics:** Power generation and buses, source switching, load shedding, hydraulic dependencies, pressure/quantity monitoring, cautions, and failure-driven loss of equipment/control authority.
- **Pressurization / air / anti-ice:** Cabin schedule, outflow control, differential-pressure limits, packs/bleeds, overheat/leak/pack failures, anti-ice loads, indications, and procedures.
- **Landing gear / brakes / ground:** Gear sequencing and indications, alternate extension, brake temperatures/failures where supported, anti-skid, autobrake, steering, and ground handling.
- **Warnings / procedures:** Consistent caution/warning logic, master lights, aural alerts, inhibit logic, checklists, non-normal procedures, and state-dependent annunciations.
- **Weight / performance:** Reliable loading and fuel state, CG limits, takeoff/landing performance, speeds, climb/cruise/descent planning, and effects of actual aircraft mass/configuration.
- **Cockpit / exterior / audio:** 747-8I-specific display presentation, readable and interactive cockpit, geometry/animations, engine/exterior model, sounds, lighting, and consistent control indications.

- **Maintenance / failures:** Failure setup, persistent-state policy, maintenance reset, test modes, system diagrams, and diagnostic logging for repeatable training scenarios.

Validation discipline

- Maintain a versioned test card per system with setup, steps, expected behavior, limits, and evidence/source.
- Use controlled conditions: one change at a time, fixed weight/CG, weather, configuration, and AP mode; capture telemetry rather than relying on memory alone.
- Test normal operation, transitions, boundaries, failures, recovery, and regression against the known-good baseline.
- Use aircraft-specific manuals or approved technical references when available. Record page/section and applicability. If unavailable, state uncertainty and avoid claiming exact Boeing fidelity.
- Keep simulation-only experimental laws separately switchable and document their limits. Never describe the add-on as certified or suitable for real-world flying.

Scope and licensing reminders

The aircraft is based on Sparky744 747-400 work. The permission exchange shared with the project identifies CC BY-NC 4.0 and invites PR contributions with README credit. The exterior-only beta is pending review and does not change internal systems. The original standalone FBW module has a scoped MIT notice; neighboring inherited/modified code and third-party components retain their own licenses. Consult the project LICENSE.md, LICENSE-MIT.md, and in-tree notices.

Guiding rule: Make the aircraft more complete by traceable system behavior, not by adding a list of impressive-sounding protections. Verify first, isolate each law, expose its state, test its failure behavior, and only then promote it to the normal aircraft configuration.

Planning document only. “Full study-level” is an aspirational project quality target; this roadmap is not Boeing documentation, engineering substantiation, or legal advice.